Finite Element Method

Finite Element Approximation: 1-D Problems I

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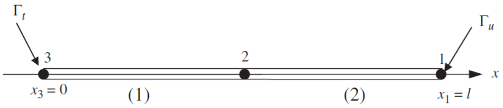
- The finite element analysis procedure is broken into four steps:
  1. **Preprocessing**: mesh construction.
  2. **Formulation** of the discrete FE equations.
  3. **Solving** the discrete equations.
  4. **Postprocessing**: display solution and compute derived quantities.
- Consider a two-linear-element model (see figure). At $x = 0$: traction (natural) BC; at $x = l$: essential BC $u(l) = \bar{u}_1$.



- The weak form (Lecture 3) is:

$$\int_0^l \left(\frac{\mathrm{d}w}{\mathrm{d}x}\right)^{\mathrm{T}} AE \left(\frac{\mathrm{d}u}{\mathrm{d}x}\right) \mathrm{d}x - \int_0^l w^{\mathrm{T}} b \, \mathrm{d}x - (w^{\mathrm{T}} \bar{t} A)\big|_{x=0} = 0 \quad (1)$$

$$\forall\, w(x) \text{ with } w(l) = 0.$$

- The FE weight functions:

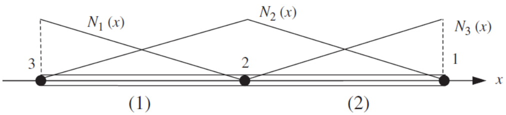
$$w(x) \approx w^h(x) = \mathbf{N}(x)\mathbf{w} \quad (2)$$

For this mesh: $w = w_1 N_1 + w_2 N_2 + w_3 N_3$.

- The FE trial solutions (Galerkin):

$$u(x) \approx u^h(x) = \mathbf{N}(x)\mathbf{d} \quad (3)$$

- Shape functions for the two-element mesh:



- The trial solutions must satisfy the essential BC. This is accomplished by setting:

$$u_1 = \bar{u}_1 \tag{4}$$

- On the essential boundary, weight functions must vanish:

$$w_1 = 0 \tag{5}$$

- Element and global matrices are related by **gather matrices**:

$$\mathbf{w}^e = \mathbf{L}^e \mathbf{w}, \qquad \mathbf{d}^e = \mathbf{L}^e \mathbf{d} \tag{6}$$

- Replace the domain integral in (1) by a sum over elements:

$$\sum_{e=1}^{n_{\mathrm{el}}} \left\{ \int_{x_1^e}^{x_2^e} \left(\frac{\mathrm{d}w^e}{\mathrm{d}x} \right)^{\mathrm{T}} A^e E^e \frac{\mathrm{d}u^e}{\mathrm{d}x} \, \mathrm{d}x - \int_{x_1^e}^{x_2^e} w^{e\mathrm{T}} b \, \mathrm{d}x - \left(w^{e\mathrm{T}} A^e \bar{t} \right) \Big|_{x=0} \right\} = 0 \tag{7}$$

- In each element e, the weight function and trial solution are written as:

$$u^e(x) = \mathbf{N}^e \mathbf{d}^e, \quad \frac{\mathrm{d}u^e}{\mathrm{d}x} = \mathbf{B}^e \mathbf{d}^e, \qquad w^{e\mathrm{T}} = (\mathbf{w}^e)^\mathrm{T} (\mathbf{N}^e)^\mathrm{T}, \quad \left(\frac{\mathrm{d}w^e}{\mathrm{d}x}\right)^\mathrm{T} = (\mathbf{w}^e)^\mathrm{T} (\mathbf{B}^e)^\mathrm{T} \tag{8}$$

- Substituting (8) into (7):

$$\sum_{e=1}^{n_{\mathrm{el}}} (\mathbf{w}^e)^\mathrm{T} \left\{ \underbrace{\int_{x_1^e}^{x_2^e} (\mathbf{B}^e)^\mathrm{T} A^e E^e \mathbf{B}^e \,\mathrm{d}x \; \mathbf{d}^e}_{\mathbf{K}^e \mathbf{d}^e} - \underbrace{\int_{x_1^e}^{x_2^e} (\mathbf{N}^e)^\mathrm{T} b \,\mathrm{d}x}_{\mathbf{f}_\Omega^e} - \underbrace{\left((\mathbf{N}^e)^\mathrm{T} A^e \bar{t} \right)_{x=0}}_{\mathbf{f}_{\Gamma^e}^e} \right\} = 0 \tag{9}$$

- Two key element matrices are defined:

(i) Element stiffness matrix

$$\mathbf{K}^e = \int_{x_1^e}^{x_2^e} (\mathbf{B}^e)^{\mathrm{T}} A^e E^e \mathbf{B}^e \,\mathrm{d}x = \int_{\Omega^e} (\mathbf{B}^e)^{\mathrm{T}} A^e E^e \mathbf{B}^e \,\mathrm{d}x \tag{10}$$

(ii) Element external force matrix

$$\mathbf{f}^e = \underbrace{\int_{\Omega^e} (\mathbf{N}^e)^{\mathrm{T}} b \,\mathrm{d}x}_{\mathbf{f}^e_\Omega} + \underbrace{\left((\mathbf{N}^e)^{\mathrm{T}} A^e \bar{t}\right)\Big|_{\Gamma^e_t}}_{\mathbf{f}^e_\Gamma} \tag{11}$$

- Substituting (10) and (11) into (9) and using $\mathbf{w}^e = \mathbf{L}^e \mathbf{w}$:

$$\mathbf{w}^{\mathrm{T}} \left(\sum_{e=1}^{n_{\mathrm{el}}} (\mathbf{L}^e)^{\mathrm{T}} \mathbf{K}^e \mathbf{L}^e \,\mathbf{d} - \sum_{e=1}^{n_{\mathrm{el}}} (\mathbf{L}^e)^{\mathrm{T}} \mathbf{f}^e \right) = 0 \tag{12}$$

- The **global stiffness matrix** and **global force vector**:

$$\mathbf{K} = \sum_{e=1}^{n_{\mathrm{el}}} (\mathbf{L}^e)^{\mathrm{T}} \mathbf{K}^e \mathbf{L}^e, \qquad \mathbf{f} = \sum_{e=1}^{n_{\mathrm{el}}} (\mathbf{L}^e)^{\mathrm{T}} \mathbf{f}^e \tag{13}$$

Element Matrices for the Two-Node Element

- From Lecture 4, the shape functions and their derivatives for the two-node element are:

$$\mathbf{N}^e = \frac{1}{l^e}\begin{bmatrix}(x_2^e - x) & (x - x_1^e)\end{bmatrix}, \qquad \mathbf{B}^e = \frac{\mathrm{d}}{\mathrm{d}x}\mathbf{N}^e = \frac{1}{l^e}\begin{bmatrix}-1 & 1\end{bmatrix}.$$

- The **element stiffness matrix** is:

$$\mathbf{K}^e = \int_{x_1^e}^{x_2^e} (\mathbf{B}^e)^{\mathrm{T}} A^e E^e \mathbf{B}^e \,\mathrm{d}x = \int_{x_1^e}^{x_2^e} \underbrace{\frac{1}{l^e}\begin{bmatrix}-1 \\ 1\end{bmatrix}}_{(\mathbf{B}^e)^{\mathrm{T}}} A^e E^e \underbrace{\frac{1}{l^e}\begin{bmatrix}-1 & 1\end{bmatrix}}_{\mathbf{B}^e} \mathrm{d}x$$

$$= \frac{A^e E^e}{(l^e)^2}\begin{bmatrix}1 & -1 \\ -1 & 1\end{bmatrix}\underbrace{(x_2^e - x_1^e)}_{l^e} = \frac{A^e E^e}{l^e}\begin{bmatrix}1 & -1 \\ -1 & 1\end{bmatrix}. \tag{14}$$

Note

This result matches the bar element stiffness derived in Lecture 2 via direct equilibrium.

- The **element body force vector** (first term of (11)), assuming linear body force $b(x) = \mathbf{N}^e\mathbf{b}$, $\mathbf{b} = [b_1 \; b_2]^{\mathrm{T}}$:

$$\mathbf{f}_\Omega^e = \int_{x_1^e}^{x_2^e} (\mathbf{N}^e)^{\mathrm{T}}\mathbf{N}^e \, \mathrm{d}x \; \mathbf{b} = \frac{1}{(l^e)^2}\int_{x_1^e}^{x_2^e} \begin{bmatrix} (x_2^e - x)^2 & (x_2^e - x)(x - x_1^e) \\ (x_2^e - x)(x - x_1^e) & (x - x_1^e)^2 \end{bmatrix} \mathrm{d}x \; \mathbf{b}$$

$$= \frac{l^e}{6}\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} b_1 \\ b_2 \end{bmatrix}. \tag{15}$$

- The global stiffness matrix is assembled from the element stiffness matrices using Eq. (13) or by direct assembly (discussed in the next lecture's example).

- Substituting (13) into (12):

$$\mathbf{w}^{\mathrm{T}}(\mathbf{Kd}-\mathbf{f})=0 \qquad \forall\, \mathbf{w} \text{ except } w_1=w(l)=0. \tag{16}$$

- Let $\mathbf{r}=\mathbf{Kd}-\mathbf{f}$. Then:

$$\mathbf{w}^{\mathrm{T}}\mathbf{r}=0 \qquad \forall\, \mathbf{w} \text{ except } w_1=0. \tag{17}$$

- This gives $w_2r_2+w_3r_3=0$ (the w_1 term drops because $w_1=0$). Since this holds for arbitrary w_2 and w_3: $r_2=r_3=0$. The value r_1 is the **reaction force** at node 1.
- Writing out the full system:

$$\mathbf{r}=\begin{bmatrix} r_1 \\ 0 \\ 0 \end{bmatrix}=\begin{bmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{bmatrix}\begin{bmatrix} \bar{u}_1 \\ u_2 \\ u_3 \end{bmatrix}-\begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} \tag{18}$$

- Rearranging (18):

$$\begin{bmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{bmatrix}\begin{bmatrix} \bar{u}_1 \\ u_2 \\ u_3 \end{bmatrix}=\begin{bmatrix} f_1+r_1 \\ f_2 \\ f_3 \end{bmatrix} \tag{19}$$

- Equation (19) is a system of 3 equations with 3 unknowns (u_2, u_3, r_1). Solution procedures were discussed in Lecture 2.
- **Partition method**: find u_2 and u_3 first, then recover r_1:

$$\begin{bmatrix} K_{22} & K_{23} \\ K_{32} & K_{33} \end{bmatrix} \begin{bmatrix} u_2 \\ u_3 \end{bmatrix} = \begin{Bmatrix} f_2 - K_{21}\bar{u}_1 \\ f_3 - K_{31}\bar{u}_1 \end{Bmatrix},$$

$$r_1 = f_1 - \begin{bmatrix} K_{11} & K_{12} & K_{13} \end{bmatrix} \begin{bmatrix} \bar{u}_1 \\ u_2 \\ u_3 \end{bmatrix}.$$

- **Direct method**: replace the essential BC row:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & K_{22} & K_{23} \\ 0 & K_{23} & K_{33} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} \bar{u}_1 \\ f_2 - K_{21}\bar{u}_1 \\ f_3 - K_{31}\bar{u}_1 \end{bmatrix},$$

$$r_1 = f_1 - \begin{bmatrix} K_{11} & K_{12} & K_{13} \end{bmatrix} \begin{bmatrix} \bar{u}_1 \\ u_2 \\ u_3 \end{bmatrix}.$$

Remark

The procedures described here apply directly to higher-order elements and to 2D/3D problems.

Next: Finite Element Approximation — 1-D Problems II

Thanks for your attention!